Atharva Sambhaji Meharkure

DS&DA Student | Python | Machine Learning | AI Systems
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SUMMARY

B.Tech student (Data science and Data Analytics) with strong fundamentals in Python, machine learning, and systems programming, plus hands-on experience with LLMs and retrieval-augmented generation (RAG). Currently building a RAG-based inventory management system; previously built an end-to-end deepfake detection pipeline, a FastAPI-backed ML system with evaluation reporting, and OS internals from scratch in C. Looking to grow into AI engineering, including model evaluation, inference, and deployment.

EDUCATION

B.Tech, Computer Science Engineering (DS) | Symbiosis Institute of Technology, Nagpur 2024 – Present
Coursework: Machine Learning, Deep Learning, NLP, Computer Vision, Data Structures & Algorithms, Probability & Statistics

TECHNICAL SKILLS

Languages: Python, C, SQL, TypeScript, Bash, x86 Assembly
ML & AI: LLMs, Retrieval-Augmented Generation (RAG), PyTorch, TensorFlow, Scikit-Learn, OpenCV, MediaPipe, LSTM
Data: Pandas, NumPy, Matplotlib
Backend & Tools: FastAPI, SQLite, pytest, Git, GitHub, Linux, React, Jupyter

PROJECTS

RAG-Based Inventory Management System | RAG, LLMs Present
- Building an inventory management system based on retrieval-augmented generation (RAG) with LLMs.

DeepNext: Deepfake Detection Engine | Python, OpenCV, MediaPipe, Scikit-Learn 2025 – Present
- Built a real-time deepfake detection system that flags synthetic media from behavioural signals, using MediaPipe's 468-point facial landmark mesh.
- Engineered behavioural biometric features (eye-blink anomaly scoring, micro-expression patterns) that feed a Scikit-Learn classifier.
- Designed the full pipeline (video in, frame extraction, face detection, feature engineering, classifier, verdict) with frame-level processing for real-time use.

Revenue Agents: Payment Recovery System | Python, FastAPI, SQLite, Scikit-Learn
- Built a Razorpay-themed demo of an autonomous payment-recovery system with a FastAPI + SQLite backend and a dashboard UI.
- Trained a Logistic Regression model for checkout routing using grouped train/test splits, and used Beta-Bernoulli Thompson Sampling to pick recovery strategies.
- Wrote an evaluation pipeline that generates interactive HTML reports, backed by a pytest suite; found and fixed a double-write bug that had inflated recovery metrics.

OS Basic Linux: Operating System Internals | C, Linux 2025 – Present
- Implemented OS fundamentals from scratch in C: processes, memory management, syscalls, file systems, IPC, and CPU scheduling.

TimesnapMRK2: Timetable Management App | TypeScript, Python, MIT License Jan 2026
- Built and open-sourced v2 of a timetable app with smart schedule structuring and conflict detection; iterated across 3 versions.

AI Tool Evaluation: Independent Research 2024 – 2025
- Evaluated 15+ AI tools (LLMs, image generators, automation platforms) using structured templates for features, limitations, and use cases; published comparisons on LinkedIn.

EXPERIENCE & LEADERSHIP

Vice Chair | IEEE 2026-2027

Coordinator | Competitive Coding Club, SIT Nagpur 2025– 2026
- Organised coding contests, workshops, and peer-learning sessions; mentored juniors in DSA fundamentals and contest preparation.
- Increased participation through structured weekly challenges and faculty coordination.